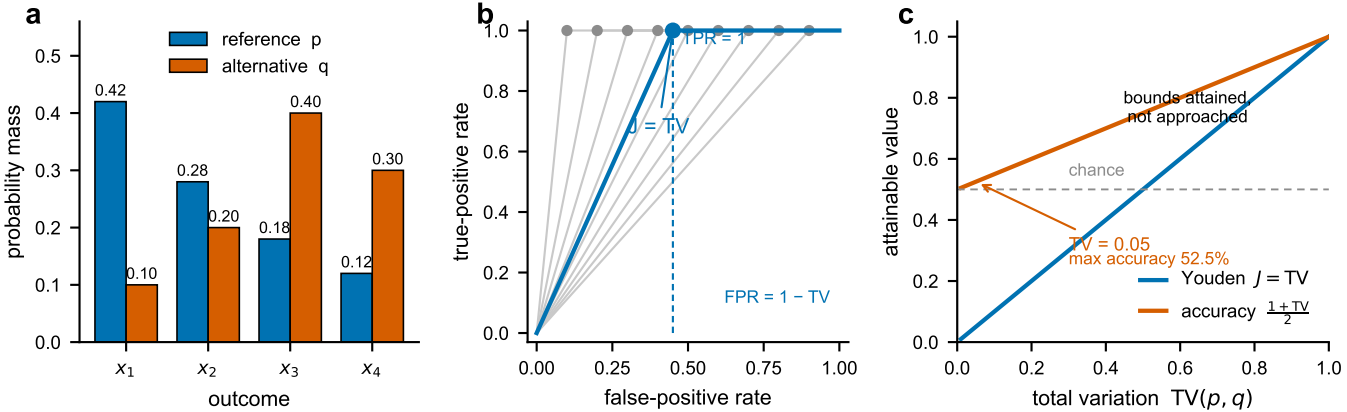
a
reference p
alternative q
probability mass
0.0
0.1
0.2
0.3
0.4
0.5
0.42
0.10
0.28
0.20
0.18
0.40
0.12
0.30
x_1
x_2
x_3
x_4
outcome
$\mathrm{TV}(p,q) = \frac{1}{2}\sum_x |p_x - q_x| = 0.40$
b
true-positive rate
false-positive rate
TPR = 1
J = TV
FPR = 1 − TV
c
attainable value
total variation TV(p, q)
bounds attained
not approached
chance
TV = 0.05
max accuracy 52.5%
Youden J = TV
accuracy $\frac{1+\mathrm{TV}}{2}$